

Cloud gaming

Cloud gaming, sometimes called **gaming on demand** or **game streaming**, is a type of online gaming that runs video games on remote servers and streams the game's output (video, sound, etc.) directly to a user's device, or more colloquially, playing a game remotely from a cloud. It contrasts with traditional means of gaming, wherein a game is run locally on a user's video game console, personal computer, or mobile device.^{[1][2]}

Background

Cloud gaming platforms operate in a similar manner to remote desktops and Photo on demand services;^[3] games are stored and executed remotely on a provider's dedicated hardware, and streamed as video to a player's device via client software. The client software handles the player's inputs, which are sent back to the server and executed in-game.^[3] Some cloud gaming services are based on access to a virtualized Windows environment, allowing users to download and install games and software as they normally would on a local computer.^{[4][5][6]}

Cloud gaming can be advantageous as it eliminates the need to purchase expensive computer hardware or install games directly onto a local game system. Cloud gaming can be made available on a wide range of computing devices, including mobile devices such as smartphones and tablets, digital media players, or proprietary thin client-like devices.^{[7][3][8]} Some services may offer additional features to take advantage of this model, including the ability for a viewer to join a player's session and temporarily take control of the game.^[9]

Due to their dependency on high-quality streaming video, cloud gaming services typically require reliable, high-speed internet connections with low latency. Even with high-speed connections available, traffic congestion and other issues affecting network latency can affect the performance of cloud gaming, and the ability to use a service regularly may also be limited by data caps enforced by some internet service providers.^{[10][11]}

Further, the costs of cloud gaming shift from traditional distribution through retail outlets and digital storefronts to the data servers that run the cloud gaming services. Novel cost structures are required to cover these operating costs compared to traditional distribution.^[7]

Infrastructure considerations

Cloud gaming requires significant infrastructure for the services to work as intended, including data centers and server farms for running the games, and high-bandwidth internet connections with low latency for delivering the streams to users.^[12] The network infrastructure required to make cloud gaming feasible was, for many years, not available in most geographic areas, or unavailable to consumer markets.^{[12][8]}

A major factor in the quality of a cloud gaming service is latency, as the amount of delay between the user's inputs and when they take effect can affect gameplay — especially in fast-paced games dependent on precise inputs (such as first-person shooters and fighting games).^{[13][14]} Attempts to reduce latency include the use of caching as the cached data can be "stored locally ... and can be retrieved when required."^[15]

The provider's dedicated hardware can be upgraded over time in order to support higher resolutions and frame rates for rendering and streams.^[7] The Quality of Experience (QoE) that measures the user's general level of satisfaction also needs to be brought into consideration during the development phase of cloud gaming.^[16]

History

Early attempts

The first demonstrated approach of cloud gaming technology was by startup G-cluster (short for Game Cluster), which introduced its product at the 2000 E3, and released around 2003. In their initial model around 2005, G-cluster provided PC games that ran on their servers, using video-on-demand service providers, set-top box manufacturers, and middleware software providers to help provide their service to network operators, and then offered the games through portals to end users. By 2010, due to changes in the market, G-cluster changed their model to work through a large server manufacturer to provide their games to the network operators and directly to users. This refocusing was necessitated by the increased available of free-to-play games available for personal computers, drawing them away from G-cluster's service, so G-cluster opted to focus on Internet Protocol television (IPTV) users instead, which had a potential target of about three million users in 2010.^[17] French telco SFR launched G-cluster gaming service in 2010 for its end users ^{[18][19]} and Orange followed suit in 2012 offering the gaming service for its customers. Both services have been commercially operational ever since offering cloud gaming for their customers on TV and mobile.

In early 2003, another attempt was announced by American company Infinium Labs, which intended to revolutionize the market of home entertainment with their Phantom video game console, a device that was envisioned to be capable of providing an on-demand video game delivery service via monthly online subscription. The Phantom was designed to run PC games, thus making it compatible with hundreds of titles from the start, and was to be sold at a much lower price than high-end PC gaming rigs. A functioning prototype was first presented at the E3 2004, running Unreal Tournament 2004, and then again at QuakeCon, where it was shown to be capable of perfectly operating Quake 3 Arena on a dedicated server. After a couple of years of setbacks, the company bankrupted in 2008, with the Phantom console never officially released to the market, becoming one of the most popular vaporware of recent times.

Video game developer Crytek began the research on a cloud gaming system in 2005 for Crysis, but according to their CEO Cevat Yerli, they halted development in 2007 to wait until the infrastructure and cable Internet providers were able to complete the task and the cost of bandwidth to decline.^[20]

OnLive and Gaikai

Entrepreneur Steve Perlman revealed OnLive at the March 2009 Game Developers Conference. Perlman stated that with improvements in data and video compression as well as the capabilities of smartphones, the potential for cloud gaming was now timely.^[21] OnLive was officially launched in June 2010, alongside the sale of its OnLive microconsole.^{[22][23]} While OnLive had acquired some support from large publishers like Ubisoft, 2K Games and THQ, they found it difficult to get other publishers on board as they were wary of the subscription price model.^[21]

Simultaneous to OnLive, another startup Gaikai was announced by David Perry in 2010. Gaikai opted to approach streaming of game demos rather than full games, making the service a form of online advertising for games. Gaikai gained far more publisher support, including Electronic Arts which OnLive had been trying to bring back to their service.^{[21][24][25]} Gaikai was acquired by Sony Computer Entertainment in July 2012 for \$340 million, and by October 2012, it was offering PlayStation games. Ultimately, the technology behind Gaikai was used as the foundation for PlayStation Now, first introduced in 2014.^{[21][26]}

OnLive was never profitable, and after a possible acquisition by HP Inc. fell through, OnLive's assets were acquired by a newly formed entity named "OL2", which was capitalized by Gary Lauder of Lauder Partners in 2012 at \$4.8 million, a fraction of OnLive's valuation from 2010. A mass layoff (2/3 of staff) was conducted to reduce operating costs.^[27] Under Lauder Partners, the new OL2 attempted to pivot its business model to allow streaming of games already owned by the user, but this failed to be profitable. OnLive and OL2's intellectual property were acquired by Sony Computer Entertainment in April 2015, but it then closed them down about a month later.^{[27][25]} As stated by The Verge, the acquisition of both Gaikai and OnLive's intellectual property gave Sony access to a range of patents covering cloud gaming.^[25]

Later advancements and adoption by the industry

Nvidia first announced its cloud gaming service, Nvidia Grid (later rebranded as GeForce Now), as a combination of hardware using its graphics processing units and software in May 2012, initially intending to partner with Gaikai for games on the service.^[28] Ubitus GameCloud was also introduced alongside Nvidia's Grid. GameCloud was designed as white-label service based on Nvidia's Grid that other providers could use to offer game streaming to their customers.^[29]

Grid was formally introduced as part of its Nvidia Shield Android TV device during the 2013 International Consumer Electronics Show. Grid/GeForce Now launched with services provided by several cloud gaming partners including Agawi, Cloudunion, Cyber Cloud, G-cluster, Playcast, and Ubitus.^[30] The Grid service was first launched in North America in November 2014 where a limited number of games were made available,^[31] and then later expanded to computers in 2017, including support for importing a user's Steam and Epic Games Store library to run on the remote instance.^{[32][33][34][35]} This importing model was criticized by publishers including Activision Blizzard and Bethesda Softworks, as purchases were only intended for personal computers and not through cloud gaming. The publishers forced NVidia to pull these games from their service.^[21]

In 2014, Dragon Quest X was brought to Nintendo 3DS in Japan using Ubitus for the streaming technology.^[36]

In 2017, the French startup Blade launched a service known as Shadow, where users are able to rent a remote Windows 10 instance on a datacenter, with allocated access to an Intel Xeon processor and Nvidia Quadro graphics. The service is geographically limited based on proximity to one of its datacenters; it initially launched in France,^[4] but began expanding into the United States in 2019.^[37]

In May 2018, Electronic Arts acquired cloud gaming assets and talent from GameFly for an undisclosed amount.^[38] EA subsequently announced "Project Atlas", a project to explore the integration of artificial intelligence, machine learning, and Frostbite engine technology to create a "unified" platform to "remotely process and stream blockbuster, multiplayer HD games with the lowest possible latency, and also to unlock even more possibilities for dynamic social and cross-platform play."^{[39][40]} That month, Google and Microsoft also announced cloud gaming initiatives, with Google beginning to pilot "Project Stream" (including a closed beta featuring Assassin's Creed Odyssey running via a client in the Google Chrome web browser,^{[41][42]} and Microsoft announced the upcoming Project xCloud, leveraging Microsoft Azure technology.^[43]

At the Game Developers Conference in 2019, Google officially announced its cloud gaming service Stadia, which officially launched on November 19 of that year.^{[44][45]} In May, Sony announced a partnership with Microsoft to co-develop cloud solutions between divisions, including gaming.^[46]

Apple Inc., which makes the iOS platform for iPhones and iPads, had looked to block cloud gaming apps on its service in mid-2020. They argued that cloud gaming services allowed developers to add games onto the iOS system that bypassed the normal checks they perform on any app before it is added to the App Store, and thus violated their terms of service.^[47] However, in September 2020, Apple altered its rules that allowed cloud gaming apps to work on iOS, with restrictions that each game must be offered as an individual download on the iOS store which the user must use before playing, though catalog apps as part of the service can list and link to these games.^[48] Both GeForce Now and Stadia announced plans in November 2020 to release iOS versions of their streaming services as progressive web applications that would be run through a Chrome or Safari browser on iOS devices, as allowed for by Apple, to support cloud gaming.^{[49][50]} Microsoft has also announced plans to use a similar approach to bring the xCloud game streaming technology to iOS via the browser sometime in early 2021.^[51]

Amazon introduced its own cloud gaming service Luna in September 2020. Games on the service will be offered via a channel-style subscription service, with Amazon's own games and those from Ubisoft available at the service's launch.^{[52][53]}

Nintendo currently has games on Nintendo Switch that primarily run on cloud gaming such as Control, Hitman 3, Marvel's Guardians of the Galaxy, and the Kingdom Hearts franchise, using Ubitus.^[54]

Future

GPU resource sharing

A proposed method to improve game streaming's scalability is adaptive graphics processing unit (GPU) resource scheduling.^[55] Most cloud gaming providers are using dedicated GPUs to each person playing a game. This leads to the best performance but can waste resources.^[55] With better GPU resource scheduling algorithms, if the game does not fully utilize that GPU it can be used to help run someone

else's game simultaneously.^[55] In the past, “GPU virtualization was not used due to the inferior performance of the resource scheduling algorithm”.^[55] However new resource management algorithms have been developed that can allow up to 90% of the GPUs original power to be utilized even while being split among many users.^[55]

Predictive input

Algorithms could be used to help predict a player's next inputs, which could overcome the impact of latency in cloud gaming applications.^[56] Stadia's head of engineering Majd Bakar foresaw the future possibility of using such a concept to "[reduce] latency to the point where it's basically nonexistent", referring to this concept as "negative latency".^[57]

See also

- 2010s in video games
- 2020s in video games
- Remote mobile virtualization for mobile gaming as a service
- Stream processing
- Real-time data

References

1. "Cloud gaming vs. console gaming: The pros and cons of each" (<https://www.digitaltrends.com/gaming/cloud-gaming-vs-console-gaming/>). *www.digitaltrends.com*. 2 October 2019. Retrieved 2019-10-28.
2. "Cloud gaming streaming services set to change your game - Latest News | Gadgets Now" (<https://web.archive.org/web/20191028225220/https://www.gadgetsnow.com/tech-news/cloud-gaming-streaming-services-set-to-change-your-game/articleshow/71771232.cms>). *Gadget Now*. 2019-10-27. Archived from the original (<https://www.gadgetsnow.com/tech-news/cloud-gaming-streaming-services-set-to-change-your-game/articleshow/71771232.cms>) on 2019-10-28. Retrieved 2019-10-28.
3. Enderle, Rob (2019-10-11). "Is xCloud – the Xbox game streaming service – a glimpse at the future of the desktop?" (<https://www.computerworld.com/article/3445366/is-xcloud-a-glimpse-at-the-future-of-the-desktop.html>). *Computerworld*. Retrieved 2020-06-04.
4. "The Shadow Ghost turns cloud gaming into a seamless experience" (<https://techcrunch.com/2019/02/27/the-shadow-ghost-turns-cloud-gaming-into-a-seamless-experience/>). *TechCrunch*. 27 February 2019. Retrieved 2020-06-04.
5. "Nvidia announces GeForce Now streaming service for PCs with pay-per-minute gaming" (<https://www.extremetech.com/gaming/242165-nvidia-announced-geforce-now-streaming-service-pcs-pay-per-minute-gaming>). *ExtremeTech*. Retrieved 5 May 2017.
6. "Nvidia brings GeForce Now game streaming to any PC or Mac" (<https://arstechnica.com/gaming/2017/01/nvidia-geforce-now-streaming-pc-mac-price-release-date/>). *Ars Technica*. 5 January 2017. Retrieved 5 May 2017.
7. Warren, Tom (2019-06-19). "What is cloud gaming? Google Stadia and Microsoft xCloud explained" (<https://www.theverge.com/2019/6/19/18683382/what-is-cloud-gaming-google-stadia-microsoft-xcloud-faq-explainer>). *The Verge*. Retrieved 2020-06-04.

8. Nelius, Joanna (2019-03-05). "What you need to know about the current state of cloud gaming" (<https://www.pcgamer.com/what-you-need-to-know-about-the-current-state-of-cloud-gaming/>). *PC Gamer*. Retrieved 2019-11-04.
9. Gilliam, Ryan (2019-03-19). "Stadia lets streamers play games with their viewers" (<https://www.polygon.com/2019/3/19/18272976/google-stadia-gdc-2019-crowd-play-youtube-streamers>). *Polygon*. Retrieved 2019-11-06.
10. "Data caps must die: How Google and Microsoft's cloud gaming ambitions could conquer ISP greed" (<https://www.pcworld.com/article/3375205/data-caps-must-die-how-google-and-microsofts-cloud-gaming-ambitions-could-conquer-isp-greed.html>). *PCWorld*. 2019-03-22. Retrieved 2020-06-04.
11. "Report: Google Stadia Cloud Gamers Poised to Exceed Internet Data Caps -" (<https://www.telecompetitor.com/report-google-stadia-cloud-gamers-poised-to-exceed-internet-data-caps/>). *Telecompetitor*. Retrieved 2020-06-04.
12. Chen, Kuan-Ta (February 4, 2014). "On the Quality of Service of Cloud Gaming Systems". *IEEE Transactions on Multimedia*. **16** (2): 480–495. doi:10.1109/TMM.2013.2291532 (<https://doi.org/10.1109%2FTMM.2013.2291532>). S2CID 6315663 (<https://api.semanticscholar.org/CorpusID:6315663>).
13. "What latency feels like on Google's Stadia cloud gaming platform" (<https://techcrunch.com/2019/03/20/what-latency-feels-like-on-googles-stadia-cloud-gaming-platform/>). *TechCrunch*. 20 March 2019. Retrieved 2020-06-04.
14. Graft, Kris (18 November 2019). "Stadia delivers on low-latency, high quality cloud gaming - but is that enough?" (https://web.archive.org/web/20191119035031/https://www.gamasutra.com/view/news/354209/Stadia_delivers_on_lowlatency_high_quality_cloud_gaming_but_is_that_enough.php). *Gamasutra.com*. Archived from the original (https://www.gamasutra.com/view/news/354209/Stadia_delivers_on_lowlatency_high_quality_cloud_gaming_but_is_that_enough.php) on November 19, 2019. Retrieved 2020-06-04.
15. "What is cloud gaming latency and the ways to reduce it" (<https://www.helpwire.app/blog/cloud-gaming-latency/>). *HelpWire Blog*. 2022-04-26. Retrieved 2022-06-20.
16. Laghari, Asif Ali; He, Hui; Memon, Kamran Ali; Laghari, Rashid Ali; Halepoto, Imtiaz Ali; Khan, Asiya (2019-10-25). "Quality of experience (QoE) in cloud gaming models: A review" (<https://www.medra.org/servlet/aliasResolver?alias=iospress&doi=10.3233/MGS-190313>). *Multiagent and Grid Systems*. **15** (3): 289–304. doi:10.3233/MGS-190313 (<https://doi.org/10.3233%2FMGS-190313>). hdl:10026.1/14894 (<https://hdl.handle.net/10026.1%2F14894>). S2CID 207959066 (<https://api.semanticscholar.org/CorpusID:207959066>).
17. Ojala, Arto; Tyrvaenen, Pasi (2011). "Developing cloud business models: A case study on cloud gaming" (<http://urn.fi/URN:NBN:fi:jyu-2011111511682>). *IEEE Software*. **28** (4): 42–47. doi:10.1109/MS.2011.51 (<https://doi.org/10.1109%2FMS.2011.51>). S2CID 14621966 (<https://api.semanticscholar.org/CorpusID:14621966>).
18. "SFR launches on-demand video games on TV" (<http://www.telecompaper.com/news/article.aspx?cid=763381>).
19. "Reportage : SFR dévoile son service de jeux vidéo "cloud gaming" sur Neufbox" (<https://www.clubic.com/connexion-internet/fai-sfr-box-neufbox/actualite-373750-cloud-gaming-sfr-service-jeux-video-neufbox.html>). 20 October 2010.
20. Martin, Matt (April 6, 2009). "Crysis Core" (<https://www.gamesindustry.biz/articles/crysis-core>). *GamesIndustry.biz*. Retrieved August 14, 2021.
21. Mangalindan, JP (October 15, 2020). "Cloud gaming's history of false starts and promising reboots" (<https://www.polygon.com/features/2020/10/15/21499273/cloud-gaming-history-onlive-stadia-google>). *Polygon*. Retrieved August 14, 2021.
22. Perlman, Steve (2010-03-10). "OnLive: Coming to a Screen Near You" (<https://web.archive.org/web/20100312043136/http://blog.onlive.com/2010/03/10/onlive-coming-to-a-screen-near-you/>). OnLive.com. Archived from the original (<http://blog.onlive.com/2010/03/10/onlive-coming-to-a-screen-near-you/>) on 2010-03-12. Retrieved 2010-03-10.

23. Shiels, Maggie (2010-03-11). "'Console killer' OnLive to launch in June" (<http://news.bbc.co.uk/2/hi/technology/8556874.stm>). *news.bbc.co.uk*. Retrieved 2010-03-11.
24. "Gaikai enters closed beta, we get an exclusive first look" (<https://www.engadget.com/2010-12-02-gaikai-enters-closed-beta-we-get-an-exclusive-first-look.html>). *Engadget*. 2 December 2010. Retrieved 2020-06-04.
25. Hollister, Sean (2019-12-05). "How Sony bought, and squandered, the future of gaming" (<https://www.theverge.com/2019/12/5/20993828/sony-playstation-now-cloud-gaming-gaikai-onlive-google-stadia-25th-anniversary>). *The Verge*. Retrieved 2020-06-04.
26. "PlayStation 4 will stream PS1, PS2, PS3 games" (<http://www.joystiq.com/2013/02/20/playstation-4-will-stream-ps1-ps2-ps3-games/>).
27. Lowensohn, Josh (2015-04-02). "Sony buys streaming games service OnLive only to shut it down" (<https://www.theverge.com/2015/4/2/8337955/sony-buys-onlive-only-to-shut-it-down>). *The Verge*. Retrieved 2020-06-04.
28. Hollister, Sean (May 15, 2012). "Nvidia announces GeForce Grid: cloud gaming direct from a GPU, with games by Gaikai" (<https://www.theverge.com/2012/5/15/3022233/nvidia-geforce-grid-gtc-2012>). *The Verge*. Retrieved August 22, 2012.
29. Holister, Sean (May 16, 2012). "Ubitus GameCloud: the white-label cloud gaming service seeking a US audience" (<https://www.theverge.com/2012/5/16/3023502/ubitus-cloud-gaming-service-gtc-2012>). *The Verge*. Retrieved August 22, 2012.
30. Gilbert, Ben (January 6, 2013). "NVIDIA details the Grid, a card built for powering cloud computing" (<https://www.engadget.com/2013-01-06-nvidia-grid-unveiled.html>). *Engadget*. Retrieved August 22, 2021.
31. Robertson, Adi (November 13, 2014). "Nvidia rolling out cloud gaming for its Shield tablet and console" (<https://www.theverge.com/2014/11/13/7211181/nvidia-launching-grid-cloud-gaming-for-shield-platform>). *The Verge*. Retrieved August 22, 2021.
32. "NVIDIA Shield TV review: the best Android set-top box you can buy" (<https://www.engadget.com/2015/06/21/nvidia-shield-tv-review/>). *Engadget*. 21 June 2015. Retrieved 4 May 2017.
33. "NVIDIA's GeForce NOW - GRID Cloud Gaming Service Goes the Subscription Way" (<http://web.archive.org/web/20151001221934/http://www.anandtech.com/show/9673/nvidias-geforce-now-grid-cloud-gaming-service-goes-the-subscription-way>). *Anandtech*. Purch, Inc. Archived from the original (<http://www.anandtech.com/show/9673/nvidias-geforce-now-grid-cloud-gaming-service-goes-the-subscription-way>) on October 1, 2015. Retrieved 4 May 2017.
34. "Nvidia finally launches GeForce Now cloud gaming for Shield set-top console" (<https://venturebeat.com/2015/09/30/nvidia-finally-launches-geforce-now-cloud-gaming-for-shield-set-top-console/>). *VentureBeat*. 30 September 2015. Retrieved 2016-07-18.
35. Warren, Tom (2018-01-08). "This app can transform your cheap laptop into a gaming PC" (<https://www.theverge.com/2018/1/8/16865480/nvidia-geforce-now-windows-beta-app-gaming-pc-ces-2018>). *The Verge*. Retrieved 2020-06-04.
36. Pereira, Chris (2014-07-08). "Dragon Quest X Uses Streaming Tech to Come to 3DS in Japan" (<https://www.gamespot.com/articles/dragon-quest-x-uses-streaming-tech-to-come-to-3ds-/1100-6420972/>). *GameSpot*. Archived (<https://web.archive.org/web/20140709230859/http://www.gamespot.com:80/articles/dragon-quest-x-uses-streaming-tech-to-come-to-3ds-/1100-6420972/>) from the original on 2014-07-09. Retrieved 2021-10-06.
37. "Cloud gaming platform Shadow brings its new plans to the US" (<https://techcrunch.com/2020/03/12/cloud-gaming-platform-shadow-brings-its-new-plans-to-the-us/>). *TechCrunch*. 12 March 2020. Retrieved 2020-06-04.
38. Musil, Steven. "EA acquires GameFly subsidiary's cloud technology assets" (<https://www.cnet.com/news/ea-acquires-gamefly-subsidiarys-cloud-technology-assets/>). *CNET*. Retrieved 2020-06-04.

39. "EA embraces game streaming with its new 'Project Atlas' engine" (<https://www.engadget.com/2018-10-30-ea-project-atlas-engine-game-streaming.html>). *Engadget*. 30 October 2018. Retrieved 2020-06-04.
40. Lanier, Liz (2018-10-30). "More Than 1,000 Employees Working on EA's Project Atlas Cloud Gaming Service" (<https://variety.com/2018/gaming/news/ea-project-atlas-cloud-gaming-1203014456/>). *Variety*. Retrieved 2020-06-04.
41. Etienne, Stefan (2018-10-08). "Google's Project Stream is a working preview of the future of game streaming" (<https://www.theverge.com/2018/10/8/17950998/google-project-stream-gaming-assassins-creed-odyssey-first-impression>). *The Verge*. Retrieved 2020-06-04.
42. Singleton, Micah (2018-10-01). "Google announces Project Stream, will let testers play Assassin's Creed Odyssey for free" (<https://www.theverge.com/2018/10/1/17924492/google-project-stream-game-assassins-creed-odyssey-free-testing>). *The Verge*. Retrieved 2020-06-04.
43. Bankhurst, Adam (October 8, 2018). "Microsoft Announces Global Game Streaming Service, Project xCloud, Beta Next Year" (<https://www.ign.com/articles/2018/10/08/microsoft-announces-global-game-streaming-service-project-xcloud-beta-next-year>). *IGN*. Retrieved June 9, 2019.
44. Hollister, Sean (June 6, 2019). "Google's Stadia game service is officially coming November: Everything you need to know" (<https://www.theverge.com/2019/6/6/18654632/google-stadia-price-release-date-games-bethesda-ea-doom-ubisoft-e3-2019>). *The Verge*. Retrieved June 6, 2019.
45. Hollister, Sean (October 15, 2019). "Google's Stadia cloud gaming service will launch on November 19th" (<https://www.theverge.com/2019/10/15/20908177/google-stadia-release-date-founders-edition-pixel-4-event>). *The Verge*. Retrieved October 15, 2019.
46. Kim, Matt (May 16, 2019). "Sony and Microsoft Set Aside Differences to Tackle Next-Gen Gaming's New Frontier: Cloud Streaming" (<https://web.archive.org/web/20200803172150/https://www.usgamer.net/articles/sony-and-microsoft-set-aside-differences>). *USGamer*. Archived from the original (<https://www.usgamer.net/articles/sony-and-microsoft-set-aside-differences>) on August 3, 2020. Retrieved May 17, 2019.
47. Gilbert, Ben (August 6, 2020). "When Microsoft's ambitious 'Netflix of gaming' service launches in September, it won't arrive on Apple devices – here's why" (<https://www.businessinsider.com/apple-explains-why-xbox-game-pass-is-not-on-iphone-2020-8>). *Business Insider*. Retrieved August 6, 2020.
48. Leswing, Kef (September 11, 2020). "Apple issues new rules for App Store that will impact streaming game services from Google and Microsoft" (<https://www.cnbc.com/2020/09/11/apple-app-store-new-rules-will-affect-google-stadia-microsoft-xcloud.html>). *CNBC*. Retrieved September 11, 2020.
49. Kerr, Chris (November 19, 2020). "Nvidia sidesteps the App Store to bring GeForce Now game streaming to iOS" (<https://www.gamedeveloper.com/business/nvidia-sidesteps-the-app-store-to-bring-geforce-now-game-streaming-to-ios>). *Gamasutra*. Retrieved November 19, 2020.
50. McAloon, Alissa (November 19, 2020). "Stadia plans to bypass App Store with web app iOS launch" (<https://www.gamedeveloper.com/game-platforms/stadia-plans-to-bypass-app-store-with-web-app-ios-launch>). *Gamasutra*. Retrieved November 19, 2020.
51. Statt, Nick (December 9, 2020). "Microsoft confirms the xCloud beta is coming to iOS and PC in spring 2021" (<https://www.theverge.com/2020/12/9/22165280/microsoft-xcloud-ios-pc-launch-spring-2021-announced>). *The Verge*. Retrieved December 9, 2020.
52. Browne, Ryan (September 24, 2020). "Cloud gaming battle heats up as Amazon enters the ring to take on Microsoft and Google" (<https://www.cnbc.com/2020/09/25/amazon-luna-will-take-on-microsoft-xbox-game-streaming-google-stadia.html>). *CNBC*. Retrieved September 25, 2020.

53. Peters, Jay (September 24, 2020). "How Amazon's Luna cloud gaming service compares to Stadia, xCloud, and GeForce Now" (<https://www.theverge.com/2020/9/25/21454917/amazon-luna-cloud-gaming-comparison-google-stadia-microsoft-xcloud-game-pass-geforce-now>). *The Verge*. Retrieved September 24, 2020.
54. Lane, Gavin (February 17, 2022). "Nintendo Switch Cloud Games" (<https://www.nintendolife.com/guides/nintendo-switch-cloud-games-list>). *NintendoLife*. Retrieved February 17, 2022.
55. Yadav, Himanshu; Annappa, B (November 2017). "Adaptive GPU resource scheduling on virtualized servers in cloud gaming". *2017 Conference on Information and Communication Technology (CICT)*. IEEE. pp. 1–6. doi:10.1109/infocomtech.2017.8340641 (<https://doi.org/10.1109%2Finfocomtech.2017.8340641>). ISBN 9781538618660. S2CID 5010275 (<https://api.semanticscholar.org/CorpusID:5010275>).
56. Lee, Kyungmin; Chu, David; Cuervo, Eduardo; Kopf, Johannes; Degtyarev, Yury; Grizan, Sergey; Wolman, Alec; Flinn, Jason (2015). "Outatime" (<http://dl.acm.org/citation.cfm?doid=2742647.2742656>). *Proceedings of the 13th Annual International Conference on Mobile Systems, Applications, and Services*. Florence, Italy: ACM Press. pp. 151–165. doi:10.1145/2742647.2742656 (<https://doi.org/10.1145%2F2742647.2742656>). ISBN 978-1-4503-3494-5. S2CID 612942 (<https://api.semanticscholar.org/CorpusID:612942>).
57. "Google Stadia Could Reach 'Negative Latency'—We'll See!" (<https://www.wired.com/story/google-stadia-negative-latency/>). *Wired*. ISSN 1059-1028 (<https://search.worldcat.org/issn/1059-1028>). Retrieved 2019-11-06.

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